

LearnFlow

****A schema-driven browser learning extension for Microsoft Edge.****

LearnFlow delivers structured learning content from bundled curriculum data files directly inside your browser. Built as a Manifest V3 Edge extension with zero dependencies.

Overview

LearnFlow is a generic curriculum player that renders learning tracks, modules, lessons, glossary terms, references, and knowledge checks from structured JSON files. The first bundled curriculum covers ****Passkeys Fundamentals****.

Key Features

- Schema-driven architecture – renders any valid curriculum pack
- Multiple learning tracks within a single curriculum
- Structured lessons with rich block types (headings, paragraphs, callouts, scenarios, comparisons, knowledge checks)
- Inline knowledge checks with scoring and feedback
- Local progress tracking with auto-save
- Per-lesson notes with debounced auto-save
- Lesson bookmarking
- Searchable glossary
- Progress dashboard with per-track breakdowns
- Resume learning from where you left off
- Keyboard navigation (arrow keys for prev/next lessons)
- Accessibility-first design (semantic HTML, ARIA landmarks, focus states, skip links)
- Local-first – all data stays on your device
- Zero external dependencies
- No build step required

Screenshots

> _Screenshots can be added here after installation._

Installation

Prerequisites

- Microsoft Edge (Chromium-based)

Steps

1. Clone or download this repository

```
``bash
git clone https://github.com/your-username/learnflow-extension.git
```

2. Open Edge and navigate to [edge://extensions/](#)
3. Enable **Developer mode** (toggle in the bottom-left or top-right)
4. Click **Load unpacked**
5. Select the [learnflow-extension/](#) folder (the folder that directly contains [manifest.json](#))
6. The LearnFlow extension icon will appear in your Edge toolbar

Usage

Popup

Click the LearnFlow icon in the toolbar to access the quick launcher:

- **Resume** — continue where you left off
- **Home** — open the main learning interface
- **Glossary** — jump directly to the glossary
- **Progress** — view your progress dashboard

Full-Page App

The main learning interface opens in a full browser tab with:

- **Sidebar navigation** — Home, Glossary, Progress, Settings
- **Home** — curriculum overview and track selection
- **Track view** — modules within a track
- **Module view** — lessons within a module with completion status
- **Lesson view** — rendered lesson content with knowledge checks, notes, and bookmarks
- **Glossary** — searchable list of all terms and definitions
- **Progress** — overall and per-track completion, bookmarks, and quiz results
- **Settings** — curriculum info, references, and progress reset

Keyboard Navigation

- **Arrow Left / Arrow Right** — navigate to previous/next lesson when viewing a lesson
- **Tab** — standard keyboard navigation through all interactive elements
- **Enter / Space** — activate buttons, cards, and options

Project Structure

```
learnflow-extension/
├─ manifest.json
├─ popup.html
├─ popup.css
├─ popup.js
├─ background/
│   └─ service-worker.js
├─ pages/
│   ├── app.html
│   └─ app.css
├─ js/
│   ├── app.js
│   ├── router.js
│   ├── state.js
│   ├── storage.js
│   ├── curriculum-loader.js
│   ├── schema-validator.js
│   ├── progress.js
│   ├── assessment.js
│   ├── ui.js
│   ├── utils.js
│   └─ renderers/
│       ├── home.js
│       ├── track.js
│       ├── module.js
│       ├── lesson.js
│       ├── glossary.js
│       ├── progress.js
│       ├── settings.js
│       ├── error.js
│       └─ blocks/
│           ├── index.js
│           ├── heading.js
│           ├── paragraph.js
│           ├── bullets.js
│           ├── numbered-list.js
│           ├── callout.js
│           ├── scenario.js
│           ├── comparison.js
│           └─ knowledge-check.js
├─ data/
│   ├── curriculum-index.json
│   └─ passkeys-v1.json
├─ assets/
│   └─ icons/
│       ├── icon-16.png
│       ├── icon-48.png
│       └─ icon-128.png
└─ README.md
```

Architecture

Two-Layer Design

Layer	Purpose
Learning Engine	Routing, rendering, progress tracking, notes, bookmarks, quiz handling, local persistence
Curriculum Pack	Structured JSON defining metadata, tracks, modules, lessons, blocks, glossary, and references

Core Principles

- **Schema-driven** — the renderer reads content block types from JSON; no hardcoded screens
- **Local-first** — all data stored in `chrome.storage.local`; nothing sent to any server
- **Zero dependencies** — vanilla HTML, CSS, and JavaScript only
- **Accessibility-first** — semantic HTML, ARIA, keyboard support, readable contrast
- **Security-conscious** — all text rendered via `textContent`; strict CSP; no external scripts

Routing

Hash-based routing with parameterized paths:

Route	View
<code>#/home</code>	Home / curriculum overview
<code>#/track/{trackId}</code>	Track detail with module list
<code>#/module/{trackId}/{moduleId}</code>	Module detail with lesson list
<code>#/lesson/{trackId}/{moduleId}/{lessonId}</code>	Lesson content viewer
<code>#/glossary</code>	Searchable glossary
<code>#/progress</code>	Progress dashboard
<code>#/settings</code>	Settings and references

Storage Schema

All data persisted in `chrome.storage.local` with versioned schema for safe upgrades:

Key	Type	Description
<code>storageSchemaVersion</code>	<code>number</code>	Current storage schema version for migrations
<code>activeCurriculumId</code>	<code>string</code>	ID of the loaded curriculum
<code>lastRoute</code>	<code>string</code>	Last visited route for resume functionality
<code>lastLessonTitle</code>	<code>string</code>	Title of last visited lesson

Key	Type	Description
activeTrackId	string	Currently active track
completedLessonIds	string[]	IDs of completed lessons
completedModuleIds	string[]	IDs of completed modules
assessmentResults	object	Knowledge check results keyed by check ID
notes	object	Notes keyed by route with text and timestamp
bookmarks	array	Bookmark objects with route, title, and timestamp

Curriculum Schema

Curriculum Pack

```
{
  "id": "string",
  "title": "string",
  "version": "string",
  "language": "string",
  "description": "string",
  "level": "string",
  "estimatedMinutes": 0,
  "author": "string",
  "tracks": [],
  "modules": [],
  "lessons": [],
  "glossary": [],
  "references": []
}
```

Supported Lesson Block Types

Type	Description
heading	Section heading with configurable level (2–6)
paragraph	Body text paragraph
bullets	Unordered list
numbered-list	Ordered list
callout	Highlighted information box
scenario	Real-world scenario description
comparison	Side-by-side column comparison

Type	Description
knowledge-check	Interactive quiz with single-select options and feedback

Bundled Curriculum

Passkeys Fundamentals v1.0.0

Track	Focus	Modules	Lessons
 Foundation	Core concepts for all learners	4	6
 User & Business	Practical guidance for daily use	3	5
 Technical	WebAuthn, cryptography, implementation	3	5
 Leadership	Strategy, policy, governance	3	3

Total: 4 tracks · 13 modules · 19 lessons · 12 knowledge checks · 18 glossary terms · 6 references

Technical Stack

Component	Technology
Extension manifest	Manifest V3
Markup	HTML5
Styling	CSS3 (custom properties, grid, flexbox)
Logic	Vanilla JavaScript (ES modules)
Storage	<code>chrome.storage.local</code>
Dependencies	None
Build step	None

Browser Compatibility

Browser	Supported
Microsoft Edge (Chromium)	☒ Primary target
Google Chrome	☒ Compatible (Manifest V3)
Brave	☒ Compatible (Manifest V3)
Firefox	☒ Not supported (different extension API)

Future Considerations

The architecture preserves room for future expansion:

- Multiple bundled curricula
 - User-importable curriculum packs
 - Transcript and media schema fields
 - Full-text search
 - Theme customization
 - Cloud sync
 - Admin analytics
 - Print/export lessons
-

Contributing

1. Fork the repository
2. Create a feature branch (`git checkout -b feature/your-feature`)
3. Commit your changes (`git commit -m 'Add your feature'`)
4. Push to the branch (`git push origin feature/your-feature`)
5. Open a Pull Request

Guidelines

- No external dependencies
 - No build tools
 - All text rendering must use `textContent` (no `innerHTML` for content)
 - Maintain accessibility standards
 - Validate curriculum JSON with the schema validator before submitting new content
-

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Acknowledgments

- [FIDO Alliance](#) — passkey standards and specifications
- [W3C WebAuthn](#) — Web Authentication API standard
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